

Problem 8, Task 2

Results

Step	Time (s)	Residual: Cross-Product	Residual: Skew Conjugation
0	1.000	8.89×10^{-16}	4.44×10^{-16}
1	2.000	1.11×10^{-15}	6.66×10^{-16}
2	3.000	4.58×10^{-16}	4.44×10^{-16}
3	4.000	1.09×10^{-15}	2.78×10^{-16}
4	5.000	4.97×10^{-16}	2.22×10^{-16}

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All residuals lie in the range $\sim 10^{-16}$, which is machine epsilon for double-precision floating-point ($\approx 2.22 \times 10^{-16}$).

Interpretation

Why doesn't the simulation prove the identity?

Numerical verification is not a mathematical proof for three fundamental reasons:

Finite sampling: We only tested 5 random (v, w, ω) triples per time step a vanishingly small sample of $\mathbb{R}^3$. The identity must hold for all vectors in $\mathbb{R}^3$ and all rotation matrices $R \in \text{SO}(3)$. A bug could exist that only manifests on inputs we didn't happen to sample.

Floating-point precision limit our residuals sit pinned at machine epsilon ($\sim 10^{-16}$) the smallest difference a double precision number can represent. We cannot distinguish between the identity being exactly true and a tiny systematic error smaller than floating-point noise.

Finite time: We sampled only 5 discrete time points. While residuals remain stable at all times, we observed only the identity at these times, not over all $t \in [0, \infty)$.

What if the residual were small but not exactly zero?

If we consistently measured a residual of 1×10^{-8} (well above machine epsilon but still small) this would suggest a real, systematic error in the composition rule, not just floating-point rounding. We would need to investigate whether MuJoCo R actually follows $\text{SO}(3)$ exactly, or if there's a bug in how we're measuring the residual. A persistent non-zero error would be evidence of a genuine problem, whereas machine-epsilon residuals are consistent with the identity being exactly true (within floating-point limits).